

Voiding Cystourethrogram (VCUG)

WARWIKI

Patient Instructions · What to expect before your bladder and urethra X-ray

A voiding cystourethrogram — or “VCUG” — is an X-ray test that takes pictures of your bladder and urethra while your bladder is filled with a special dye and while you urinate. It shows how your bladder fills and empties and helps your care team find problems such as urine flowing backward toward the kidneys, a narrowing, or a leak.

About This Test

Your bladder stores urine, and the urethra is the tube that carries it out of your body. A VCUG fills the bladder with a dye that shows up white on X-ray, then records pictures while you urinate. Because it captures the bladder and urethra *in action*, it can show things a still picture cannot.

Your care team may order a VCUG to:

- See how your bladder fills and empties
- Check whether urine flows backward toward the kidneys (reflux)
- Look at the bladder neck and the urethra during urination
- Find a narrowing, a leak, or a small pouch (diverticulum)
- Check healing after bladder, prostate, or urethral surgery

The test looks at your lower urinary tract — the bladder and urethra. It is a picture that guides your care, not a treatment.

Is It Safe?

Yes. A VCUG is a common, low-risk test. The amount of X-ray used is small, and the dye leaves your body in your urine. The most common issue is a mild urinary tract infection from the catheter — this is why a clean technique is used and an antibiotic is sometimes given. Brief discomfort, mild burning, or a little blood when you urinate afterward are usually minor and pass quickly.

LEARN THE TERMS

Bladder

The organ that stores urine until you are ready to empty it.

Urethra

The tube that carries urine from the bladder out of the body.

Catheter

A thin, soft tube passed through the urethra into the bladder. For this test it fills the bladder with dye.

Contrast dye

A liquid that shows up white on X-ray so the bladder and urethra can be seen.

Voiding

Urinating — emptying your bladder. The key pictures are taken while you void.

Reflux

When urine flows backward from the bladder toward the kidneys. A VCUG can show this.

Fluoroscopy

A type of X-ray that shows live, moving pictures as the dye flows.

Post-void image

A picture taken after you urinate, to see how well your bladder emptied.

WILL IT HURT? Placing the catheter may cause brief pressure or stinging, and a full bladder feels like a strong urge to urinate. Being asked to urinate during the test is normal — the team does this every day and keeps you covered. Tell them if you are uncomfortable. The test takes about 30 minutes.

How to Get Ready (Before Your Test)

There is very little to do to prepare. In most cases:

- You can **eat and drink normally** — no need to fast unless your team tells you otherwise.
- Take your **usual medicines** unless your team asks you to stop something.
- You can usually **drive yourself home**, because no sedation is needed for most adults.

Tell your team ahead of time if you:

- Have ever had a reaction to X-ray contrast dye or iodine
- Take a blood thinner
- **Could be pregnant** — X-rays are usually avoided during pregnancy
- Have signs of a urine infection (burning, fever, or cloudy, strong-smelling urine) — an active infection may mean the test is rescheduled

You may be asked to give a **urine sample** to check for infection, and you may be given an **antibiotic** to lower the small risk of infection.

What Happens During the Test

- 1 You lie on an X-ray table, and the opening where urine comes out is cleaned.
- 2 A thin, soft catheter is gently passed through the urethra **into your bladder**. This may cause brief pressure or stinging.
- 3 Your bladder is slowly filled with dye through the catheter. You will feel growing fullness, like needing to urinate. Pictures are taken as it fills.
- 4 When your bladder is full, you will be asked to **urinate while the X-ray records** — usually with the catheter removed. This is the most important part of the test.
- 5 A final picture may be taken after you empty, to see how well your bladder emptied.

After the Test

- You can usually go home and return to **normal activities right away**.
- You may feel **mild burning when you urinate** for a day or so. Drinking extra water helps flush it out and eases this.
- You may notice a **small amount of blood**, or a pink tinge in your urine. This is common and usually clears within a day.
- The dye leaves your body in your urine over the next several hours.

Call your care team or seek care if you have:

- A fever or chills
- Trouble urinating, or you cannot urinate at all
- Heavy bleeding, or bleeding that will not stop
- Pain that gets worse instead of better

THREE THINGS TO REMEMBER

1. A VCUG is a safe X-ray that fills your bladder with dye and films it while you urinate — it shows how your bladder and urethra work and finds problems like backflow toward the kidneys.
2. Little to prepare: eat, drink, and take your usual medicines — but tell your team about any dye allergy, blood thinners, a chance of pregnancy, or signs of a urine infection.
3. A catheter is placed and you will be asked to urinate during the test — this is normal. Mild burning or a little blood for a day is expected; drink extra water, and call for fever, heavy bleeding, or trouble urinating.